

Combined Oral Contraceptives and Stroke Risk in Individuals With Migraine With Aura

Liza R. Gibbs, PhD, MS, Matthew P. Fox, DSc, MPH, Hugo J. Aparicio, MD, MPH, and Susan Jick, DSc, MPH

OBJECTIVE: To estimate the effect of combined oral contraceptives (COCs) compared with non-estrogen-containing oral contraceptives on the risk of stroke among individuals with migraine with aura.

METHODS: Using U.K. primary care medical records with linked hospital data from the Clinical Practice Research Datalink, the authors evaluated the risk of stroke associated with use of COCs, containing estrogen and progestogen, compared with progestogen-only pills (POPs) among individuals with migraine with aura. The study included individuals aged 15–49 years who initiated either contraceptive of interest from 2000 to 2024, with no history of stroke, other thrombotic disease, or menopause at the time of initiation. Risk of ischemic and any stroke was compared after application of treatment and censoring weights using risk differences and ratios from the Kaplan-Meier estimator. Indi-

viduals were censored on medication discontinuation in primary analyses.

RESULTS: Among the 12,429 initiators of COCs and 41,984 initiators of POPs included in the study, 1-year risk of ischemic stroke was 17.36 and 17.62 per 100,000 people, respectively. After weighting, the 1-year risk difference for ischemic stroke for COCs compared with POPs was 9.53 per 100,000 people (95% CI, –38.65 to 57.71). For stroke of any type, the risk difference was 3.49 per 100,000 people (95% CI, –45.17 to 52.16). Effects were larger with shorter follow-up periods (3-month ischemic stroke risk difference of 17.62/100,000 people) but were still imprecisely measured. Findings were similar when individuals were not censored at medication discontinuation (ischemic stroke intent-to-treat risk difference of 10.36/100,000 people).

CONCLUSION: The risk of stroke in this reproductive-age population of individuals with migraine with aura was low overall and similar between users of COCs and POPs.

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See related editorial on page 157.

From the Department of Epidemiology, Boston University School of Public Health, and the Department of Neurology, Boston University Chobanian & Avedisian School of Medicine, Boston, and the Boston Collaborative Drug Surveillance Program, Lexington, Massachusetts; and Aetion, a Datavant Company, New York, New York.

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Corresponding author: Liza R. Gibbs, PhD, MS, Department of Epidemiology, Boston University School of Public Health, Boston, MA; lgibbs@bu.edu.

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Approximately 12% of women of reproductive age in the United Kingdom and 14% in the United States use oral contraceptives.^{1,2} Oral contraception pill types include combined oral contraceptives (COCs) containing estrogen and progestogen, as well as progestogen-only pills (POPs). Choice of contraception may be influenced by a variety of medical and personal factors including childbearing goals, socioeconomic status, access to care, ease of use (ie, convenience), effectiveness for contraception or treatment of other conditions, safety, and side effects.³

Migraine with aura is a reversible focal neurologic phenomenon typically experienced as a set of visual or sensory symptoms that accompany or precede a migraine headache.⁴ More than 40% of women will experience migraine of any type over their lifetimes, with up to one-third experiencing migraine with aura.^{4,5} Use of COCs is contraindicated in people

with migraine with aura because of the elevated stroke risk associated with these contraceptives and this type of migraine.^{6–8} This contraindication is largely supported by older studies of higher-dose formulations (eg, 50 micrograms ethinyl estradiol or more) than those commonly prescribed today.^{9,10} Further, most relevant research conducted to date has measured relative stroke risks conferred by the presence of migraine with adjustment, stratification, or restriction by oral contraceptive use,^{11–16} or the joint effect of migraine with aura and oral contraceptive use compared with the absence of both conditions,¹⁷ rather than assessing modern contraceptive options among individuals with migraine with aura. Importantly, the absolute risk of stroke among reproductive-age women is low even in the presence of both migraine with aura and COC use (1-year risk of 15/100,000 women aged 20–44 years).¹⁸

More robust research on the effect of modern hormonal contraceptives in the presence of migraine with aura is needed to inform clinical decision-making regarding choice of contraception in this population.^{9,10} This study aims to estimate the effect of initiation and use of COCs compared with non-estrogen-containing oral contraceptives on the risk of stroke among female individuals aged 15–49 years with migraine with aura.

METHODS

This retrospective cohort study was conducted using the Clinical Practice Research Datalink Aurum, a longitudinal U.K. population-based, anonymized electronic medical record database. The Clinical Practice Research Datalink contains records for more than 19 million patients, including more than 7 million patients treated in currently active medical practices, primarily residing in England.¹⁹ Data in the Clinical Practice Research Datalink include demographic and patient characteristics, clinical diagnoses, and prescribed medications. Validation studies have found Clinical Practice Research Datalink Aurum to be of high quality and completeness suitable for observational research.²⁰ This study used records captured predominantly from 1995 to September 2024, with some additional records captured more sparsely before 1995. Eligibility criteria, exposure status, baseline covariates, and outcomes were evaluated in Clinical Practice Research Datalink Aurum. Linked Hospital Episode Statistics data, available for approximately three quarters of individuals, were additionally leveraged for evaluation of outcomes. This study was approved by the Clinical Practice Research Data-

link Research Data Governance (reference 25_005087).

This study population comprised female individuals of reproductive age with migraine with aura who received primary care at practices in the United Kingdom captured in Clinical Practice Research Datalink Aurum. Individuals initiating COCs or POPs on or after January 1, 2000, were identified according to the following eligibility criteria. *Initiators* were defined as those with no use in the past year, and as such, included both new users and reinitiators who stopped therapy 1 or more years before reinitiation. Contraceptive use was identified through prescription entries in the primary care record, which provide formulation-specific medication product codes. The *index date* was defined as the date of qualifying treatment initiation.

Eligible individuals were aged 15–49 years at the index date, had sex listed as female, had a recorded diagnosis of migraine with aura before or on the index date, and had at least 1 year of medical record history before the index date to determine new contraceptive use and assess eligibility criteria and covariates. Individuals were excluded if they had any history of stroke, indicators of other thrombotic disease or hypercoagulability (antithrombotic medication use, record of thrombogenic mutation, venous thromboembolism, transient ischemic attack, ischemic or valvular heart disease, cancer except nonmelanoma skin cancer, atrial fibrillation or flutter, or sickle cell anemia), or documented onset of menopause or perimenopause. Individuals who had used any other estrogen-containing medication in the prior year were also excluded (Appendix 1, available online at <http://links.lww.com/AOG/E658>).

The primary outcome under study was ischemic stroke. Secondary outcomes included any stroke (ischemic, hemorrhagic, or unspecified) and hemorrhagic stroke. Stroke was identified through linked hospital records (using International Classification of Diseases, Tenth Revision diagnosis codes) and from primary care records (using medical diagnosis codes comprising multiple coding systems including Read and SNOMED) (Appendix 2, available online at <http://links.lww.com/AOG/E658>). Strokes preceded or followed within 30 days by a diagnosis code for traumatic stroke were excluded. Stroke is expected to have high capture in these data through their linkage to hospital records and because of incentivized stroke reporting in primary care records through the National Health Service Quality and Outcomes Framework.

Baseline covariates used for confounding adjustment comprised factors that could plausibly affect

both contraceptive prescribing and risk of stroke. Covariates were identified based on diagnosis and prescription histories from the primary care record as of the index date and included age, year of index date, race, body mass index (BMI, calculated as weight in kilograms divided by height in meters squared), smoking status, 1-year pregnancy history, hypertension, diabetes type 1 or 2, and both acute and preventive migraine medication use (as proxies for migraine frequency). History of pregnancy and migraine medication use were evaluated in the year before and including the index date, and comorbidities were evaluated in all available record history before or on the index date. Smoking status and BMI were reported as the most recent value recorded before or on the index date. The dose of estrogen at index (in the COC group) and availability of linked hospital data were described but not adjusted for.

Treatment groups were balanced at baseline by stabilized inverse probability of treatment weights estimated based on selected study covariates to control for anticipated confounding in these data (eg, by indication), with absolute standardized differences less than 0.1 between groups considered balanced. For covariates with missingness (BMI, smoking status, and race), missing values were reported descriptively and imputed by multiple imputation with five iterations using the MICE algorithm for weighting in comparative analyses, pooled according to Rubin's rules.^{21,22}

Individuals were followed starting on the day after the index date using an as-treated approach (censoring at discontinuation of or switching from index treatment, end of available data, death, or outcome) as the primary analysis. To account for expected continued elevation in clotting factors after COC discontinuation²³ and potential differences between the prescription date and the date medication was taken, the date of discontinuation for both groups was defined as 90 days after the end of supply of the last prescription received before a gap of 90 days or more between prescription end and any subsequent restart. To account for differential censoring by exposure group, stabilized inverse probability of censoring weights estimated using baseline covariates were additionally applied in as-treated analyses.

Outcomes were compared using risk difference and risk ratio at 1 year using the Kaplan-Meier estimator; 95% CIs were estimated using survival probability variances calculated by Greenwood's²⁴ formula. In secondary analyses, risks were additionally compared at 3, 6, and 18 months of follow-up.

An intention-to-treat sensitivity analysis (without censoring at discontinuation or switching of index medication) was performed given the greater risk of differential censoring posed by the primary as-treated analysis. Sensitivity analyses were also conducted among the subset of individuals with hospital data linkage to evaluate any potential differences in linkage by exposure status, and after exclusion of those taking non-ethinyl estradiol COCs. All analyses were performed using R 4.3.3.

RESULTS

A total of 54,413 female individuals with migraine with aura had a qualifying incident prescription for oral contraceptives and met all study eligibility criteria. Of these, 12,429 (22.8%) had a prescription for COCs, and 41,984 (77.2%) had a prescription for POPs (Appendix 3, available online at <http://links.lww.com/AOG/E658>).

Compared with those prescribed POPs, individuals prescribed COCs were generally younger (median age 24 years vs 27 years), had earlier index dates (42.7% before 2010 vs 12.0%), and had less history of recent pregnancy (20.2% vs 24.8%) or migraine treatments (14.9% vs 23.6%). Those prescribed COCs had more missing data for race, smoking status, and BMI. The two groups had similar rates of hospital data linkage (73.3% vs 71.6%) and similarly low prevalences of hypertension and diabetes (less than 2%) (Table 1). Among those prescribed COCs, the most common dose of ethinyl estradiol was 30 micrograms (71.4%), followed by 35 micrograms (16.7%), 20 micrograms (9.7%), and more than 35 micrograms (1.9%); 24 individuals were prescribed other types of estrogen (0.2%).

After application of treatment weights, the treatment groups were similar with respect to all measured baseline characteristics. After incorporation of imputed values, the majority of the weighted population was White (89.8–90.0%), most were nonsmokers (59.2–60.1%), and more than half had normal BMI (52.2–54.0%) (Table 1).

In as-treated analyses, six individuals had an ischemic stroke across 1 year of follow-up (two in the COC group and four in the POP group), equating to a 1-year risk per 100,000 people of 17.36 (95% CI, 0.00–41.44) and 17.62 (95% CI, 0.00–36.48), respectively, when accounting for censoring. Only one individual, in the COC group, had a hemorrhagic stroke. Stroke of any type occurred in nine individuals (two in the COC group and seven in the POP group); one individual had both an ischemic and a hemorrhagic stroke. In analyses using intent-to-treat censoring

Table 1. Demographics and Baseline Characteristics by Contraceptive Exposure Group Among Female Individuals With Migraine With Aura in the United Kingdom

Characteristic	Before Weighting (Unadjusted)			After Weighting* (Adjusted)		
	COC Group (n=12,429)	POP Group (n=41,984)	ASD	COC Group (n=12,036.6)	POP Group (n=41,878.6)	ASD
Demographics						
Age at index date (y)			0.28			0.04
Mean $\pm$ SD	25.4 $\pm$ 7.5	27.6 $\pm$ 8.3		26.9 $\pm$ 7.9	27.2 $\pm$ 8.2	
Median [IQR]	24 [19, 31]	27 [21, 33]		26 [20, 33]	26 [20, 33]	
Race			0.21			0.01
Asian	409 (3.3)	1,052 (2.5)		355.8 (3.0)	1,236.3 (3.0)	
Black	153 (1.2)	654 (1.6)		179.0 (1.5)	679.6 (1.6)	
White	9,413 (75.7)	34,413 (82.0)		10,832.9 (90.0)	37,595.1 (89.8)	
None of the above	566 (4.6)	2,220 (5.3)		668.9 (5.6)	2,367.7 (5.7)	
Missing	1,888 (15.2)	3,645 (8.7)		0 (0)	0 (0)	
Year of index date			0.86			0.04
2000–2004	2,621 (21.1)	1,527 (3.6)		951.1 (7.9)	3,103.4 (7.4)	
2005–2009	2,696 (21.7)	3,532 (8.4)		1,429.3 (11.9)	4,800.9 (11.5)	
2010–2014	2,840 (22.8)	8,009 (19.1)		2,443.5 (20.3)	8,360.0 (20.0)	
2015–2019	2,640 (21.2)	14,189 (32.8)		3,770.1 (31.3)	12,987.6 (31.0)	
2020–2024	1,632 (13.1)	14,727 (35.1)		3,442.6 (28.6)	12,626.6 (30.2)	
Hospital data linkage	9,110 (73.3)	30,056 (71.6)	0.04	8,587.1 (71.3)	30,195.9 (72.1)	0.02
Clinical characteristics and comorbidities						
Smoking status			0.14			0.02
Current	2,751 (22.1)	9,160 (21.8)		2,711.3 (22.5)	9,662.0 (23.1)	
Former	1,883 (15.2)	7,567 (18.0)		2,094.8 (17.4)	7,443.9 (17.8)	
Nonsmoker	6,812 (54.8)	23,192 (55.2)		7,230.5 (60.1)	24,772.7 (59.2)	
Missing	983 (7.9)	2,065 (4.9)		0 (0)	0 (0)	
BMI (kg/m ²)			0.33			0.07
Lower than 18.5 (underweight)	807 (6.5)	2,625 (6.3)		965.8 (8.0)	3,373.0 (8.1)	
18.5–24.9 (normal)	5,420 (43.6)	18,642 (44.4)		6,504.6 (54.0)	21,858.3 (52.2)	
25–29.9 (overweight)	2,269 (18.3)	8,632 (20.6)		2,782.8 (23.1)	9,607.2 (22.9)	
30–39.9 (obesity)	1,076 (8.7)	5,964 (14.2)		1,580.5 (13.1)	6,039.5 (14.4)	
40 or higher (severe obesity)	113 (0.9)	1,090 (2.6)		203.0 (1.7)	1,000.6 (2.4)	
Missing	2,744 (22.1)	5,031 (12.0)		0 (0)	0 (0)	
Hypertension	67 (0.5)	614 (1.5)	0.09	108.2 (0.9)	524.0 (1.3)	0.03
Diabetes type 1 or 2	61 (0.5)	337 (0.8)	0.04	76.5 (0.6)	314.2 (0.8)	0.01
Clinical history in previous year						
Pregnancy	2,516 (20.2)	10,412 (24.8)	0.11	2,886.1 (24.0)	10,030.5 (24.0)	0.00
Migraine treatment						
Preventive	1,035 (8.3)	6,125 (14.6)	0.20	1,413.7 (11.7)	5,517.9 (13.2)	0.04
Acute	1,107 (8.9)	6,030 (14.4)	0.17	1,428.5 (11.9)	5,507.1 (13.2)	0.04
Any	1,851 (14.9)	9,922 (23.6)	0.22	2,360.8 (19.6)	9,090.8 (21.7)	0.05

COC, combined oral contraceptive; POP, progestogen-only pill; ASD, absolute standardized difference; BMI, body mass index.

Data are mean $\pm$ SD, median [IQR], or n (%) unless otherwise specified.

Bold indicates ASD values of 0.1 or greater (meaningful differences).

* The weighted population is shown after application of stabilized inverse probability of treatment weights estimated based on age, year of index date, race, BMI, smoking status, 1-year pregnancy history, hypertension, diabetes type 1 or 2, acute migraine medication use in the previous year, and preventive migraine medication use in the previous year. Results for the first imputation are reported.

without censoring on medication discontinuation or switching, an additional two ischemic strokes were observed in the POP group. Because only one hemorrhagic stroke was observed, comparative results are reported only for ischemic stroke and for any stroke

(Table 2, Appendix 4, available online at <http://links.lww.com/AOG/E658>).

The median follow-up time in as-treated analyses was 256–258 days; approximately half of individuals were censored due to contraceptive discontinuation

Table 2. One-Year Risk of Ischemic or Any Stroke Comparing Users of Combined Oral Contraceptives With Users of Progestogen-Only Pills Among Female Individuals With Migraine With Aura in the United Kingdom: Primary Outcome, Secondary Outcome, and Sensitivity Analyses

Exposure Group	Before Weighting (Unadjusted)					After Weighting* (Adjusted)		
	No. of People	No. of Events	No. of People Censored [†]	Risk/100,000 People	Crude Risk Difference/100,000 People (95% CI)	Crude RR (95% CI)	Adjusted Risk Difference/100,000 People (95% CI)	Adjusted RR (95% CI)
Primary analysis: primary and secondary outcomes								
Ischemic stroke								
COC	12,429	2	7,824	17.36	−0.26 (−30.85 to 30.33)	0.99 (0.17–5.68)	9.53 (−38.65 to 57.71)	1.47 (0.24–8.90)
POP	41,984	4	28,486	17.62	Ref	Ref	Ref	Ref
Any stroke [‡]								
COC	12,429	2	7,824	17.36	−7.87 (−39.66 to 23.91)	0.69 (0.14–3.45)	3.49 (−45.17 to 52.16)	1.13 (0.21–6.13)
POP	41,984	7	28,483	25.24	Ref	Ref	Ref	Ref
Sensitivity analysis: intent-to-treat								
Ischemic stroke								
COC	12,429	2	970	16.42	0.79 (−25.20 to 26.77)	1.05 (0.21–5.21)	10.36 (−32.70 to 53.42)	1.60 (0.29–8.77)
POP	41,984	6	5,785	15.64	Ref	Ref	Ref	Ref
Any stroke [‡]								
COC	12,429	2	970	16.42	−6.55 (−33.82 to 20.72)	0.71 (0.15–3.31)	1.93 (−42.38 to 46.23)	1.08 (0.21–5.48)
POP	41,984	9	5,785	22.97	Ref	Ref	Ref	Ref
Sensitivity analysis: restriction to those with hospital linkage								
Ischemic stroke								
COC	9,110	2	5,733	23.65	7.56 (−31.07 to 46.18)	1.47 (0.22–9.61)	20.46 (−45.27 to 86.18)	1.98 (0.29–13.54)
POP	30,056	3	20,353	16.10	Ref	Ref	Ref	Ref
Any stroke [‡]								
COC	9,110	2	5,733	23.65	3.98 (−35.27 to 43.24)	1.20 (0.21–7.04)	17.67 (−48.28 to 83.63)	1.75 (0.28–11.05)
POP	30,056	4	20,352	19.67	Ref	Ref	Ref	Ref
Sensitivity analysis: restriction to those prescribed COCs containing ethinyl estradiol								
Ischemic stroke								
COC	12,405	2	7,808	17.39	−0.23 (−30.85 to 30.39)	0.99 (0.17–5.69)	9.54 (−38.61 to 57.69)	1.47 (0.24–8.91)
POP	41,984	4	28,486	17.62	Ref	Ref	Ref	Ref

(continued)

Table 2. One-Year Risk of Ischemic or Any Stroke Comparing Users of Combined Oral Contraceptives With Users of Progestogen-Only Pills Among Female Individuals With Migraine With Aura in the United Kingdom: Primary Outcome, Secondary Outcome, and Sensitivity Analyses (continued)

Exposure Group	Before Weighting (Unadjusted)					After Weighting* (Adjusted)		
	No. of People	No. of Events	No. of People Censored [†]	Risk/100,000 People	Crude Risk Difference/100,000 People (95% CI)	Crude RR (95% CI)	Adjusted Risk Difference/100,000 People (95% CI)	Adjusted RR (95% CI)
Any stroke [‡]								
COC	12,405	2	7,808	17.39	−7.84 (−39.66 to 23.97)	0.69 (0.14–3.46)	3.51 (−45.12 to 52.14)	1.13 (0.21–6.13)
POP	41,984	7	28,483	25.24	Ref	Ref	Ref	Ref

RR, risk ratio; COC, combined oral contraceptive; POP, progestogen-only pill; Ref, reference.

* All weighted analyses were weighted by inverse probability of treatment. As-treated analyses (including the primary analysis) were additionally weighted by inverse probability of censoring weights. All weights were stabilized, and variables assessed in both weighting models included age, year of index date, race, body mass index, smoking status, 1-year pregnancy history, hypertension, diabetes type 1 or 2, acute migraine medication use in the previous year, and preventive migraine medication use in the previous year (with censoring weight models additionally including exposure status).

[†] In as-treated analyses (including the primary analysis), individuals were censored on the occurrence of stroke, end of available data, maximum follow-up (1 year), death, discontinuation of index medication, or switching from index medication. In intent-to-treat sensitivity analyses, individuals were not censored on discontinuation or switching.

[‡] Secondary outcome; includes ischemic, hemorrhagic, and unspecified stroke.

(47.7–55.2%). The median follow-up time in intent-to-treat analyses was 1 year (Appendix 5, available online at <http://links.lww.com/AOG/E658>).

After weighting, the 1-year risk difference for ischemic stroke comparing users of COCs with users of POPs was 9.53 per 100,000 people (95% CI, −38.65 to 57.71), and similarly, the risk ratio was 1.47 (95% CI, 0.24–8.90). The 1-year risk difference for any stroke was lower at 3.49 per 100,000 people (95% CI, −45.17 to 52.16), and the risk ratio was 1.13 (95% CI, 0.21–6.13) (Table 2, Fig. 1). Risk differences and ratios were higher for shorter follow-up periods, with a 3-month risk difference of 17.62 per 100,000 people (95% CI, −21.07 to 56.30) and risk ratio of 9.73 (95% CI, 0.61–155.47). All estimates were imprecisely measured, with CIs encompassing both harmful and protective effects (Table 3, Fig. 2).

In intent-to-treat analyses, results were similar. After weighting, the 1-year risk difference for ischemic stroke was 10.36 per 100,000 people (95% CI, −32.70 to 53.42), and the risk ratio was 1.60 (0.29–8.77). After restriction to individuals with linked hospital data, effect size was slightly increased, though with decreased precision: The weighted 1-year risk difference for ischemic stroke was 20.46 per 100,000 people (95% CI, −45.27 to 86.18), and the risk ratio was 1.98 (95% CI, 0.29–13.54). Exclusion of the 24 individuals with non-ethinyl estradiol COCs had a negligible influence on study findings, with an adjusted 1-year

risk difference for ischemic stroke of 9.54 per 100,000 people (95% CI, −38.61 to 57.69) and risk ratio of 1.47 (95% CI, 0.24–8.91) (Table 2).

DISCUSSION

The findings of this cohort study suggest similar 1-year risks of ischemic stroke among users of COCs and POPs, with a risk difference of 9.53 per 100,000 people comparing users of COCs with users of POPs. Risks were higher in the first 3–6 months of medication use and were attenuated with longer follow-up. Results were similar when the outcome was expanded to include strokes of any type, and when individuals were not censored on discontinuation or when switching from their index contraceptive. Due to rarity of the outcome, CIs for all estimates were wide and inclusive of the null (not statistically significant).

The overall 1-year risk of ischemic stroke in this study was 17–18 per 100,000 individuals, or up to 25 per 100,000 individuals when including strokes of any type. These risks are similar to meta-analysis-based estimates of 15 and 6 ischemic strokes per year per 100,000 women aged 20–44 years with migraine with aura, with or without use of hormonal contraceptives, respectively.¹⁸ A study of a general population of Danish women aged 15–49 years estimated risks of ischemic stroke of 39 and 33 per 100,000 person-years among those using COCs (most commonly 30–40 micrograms) and POPs, respectively.²⁵ Given that absolute risks increase with age,¹⁷ comparison

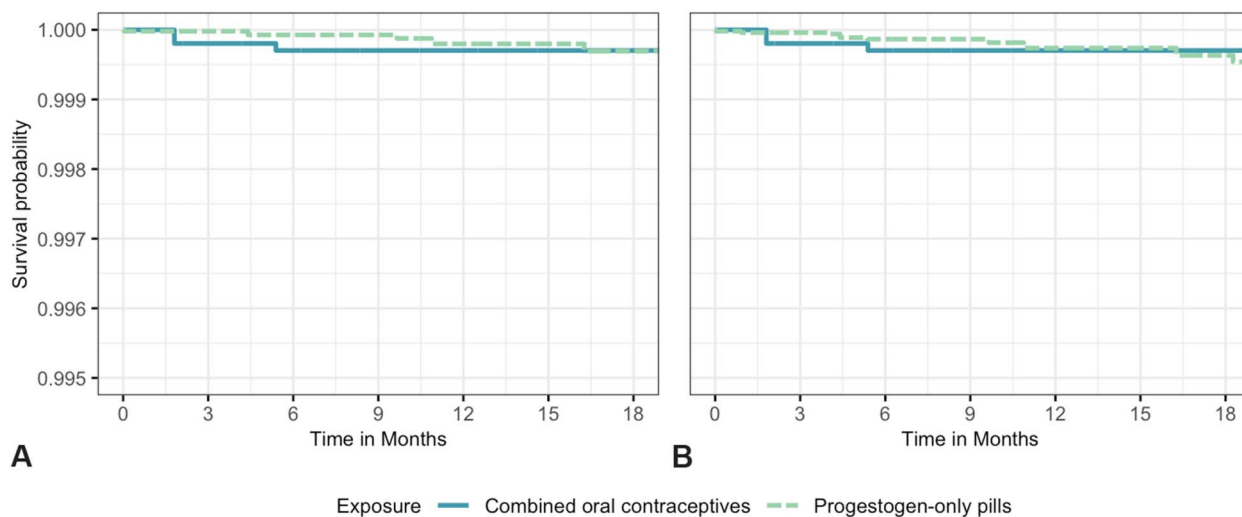


Fig. 1. Time to stroke comparing users of combined oral contraceptives with users of progestogen-only pills among female individuals with migraine with aura in the United Kingdom, after application of inverse probability of treatment and censoring weights. Ischemic stroke (**A**) and any stroke (**B**). Individuals were followed in an as-treated-style analysis from the day after contraceptive initiation to the earliest of the occurrence of stroke, end of available data, maximum follow-up, death, or discontinuation of or switching from index medication. Survival curves for the first imputation are shown. Any stroke (secondary outcome) includes ischemic, hemorrhagic, and unspecified stroke.

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across study populations is sensitive to differences in age distributions. Additionally, the present study excluded the 4% of individuals with a history of stroke or other thrombotic disease, which is expected to reduce risk of stroke in this sample compared with more general populations.

The estimates observed in the study population are similar or lower relative to estimates in general populations. In a large national Danish study from 1996 to 2021, use of COCs was associated with a statistically significant twofold increase in risk of ischemic stroke compared with nonuse, equivalent to an excess of 21 strokes per 100,000 women of reproductive age per year.²⁵ The present study observed a slightly smaller, nonstatistically significant increase in risk, with a point estimate equivalent to a 47% increase in risk of ischemic stroke or an excess of approximately 10 strokes per 100,000 individuals per year. A U.K. Biobank study of women recruited from 2006 to 2010 estimated a 2.5-fold increase in risk of stroke of any type in the first year of any oral contraceptive use, and a null effect for remaining years of use.²⁶ The present study also observed a time-varying effect; risk was highest in the 3 to 6 months after contraceptive initiation and attenuated with longer follow-up.

A strength of this study includes the use of longitudinal medical record data from the United Kingdom in which records are predominantly captured prospectively at the point of care, as well as

linked hospital data. These data contained diagnoses, detailed prescription medication information, and important individual characteristics such as smoking status and BMI. Because individuals are registered with a single primary care practice that acts as a gatekeeper to specialist care,²⁷ individuals may be followed for years or decades, with a median of 7 years of observability in the medical record preindex and 13 years postindex. Data used in this study spanned 2000 to 2024, allowing for the study of modern-day contraceptives. Nearly all COC users (98%) were prescribed doses of 35 micrograms or less of ethinyl estradiol, most commonly 30 micrograms. Conversely, many historic studies of oral contraceptives in migraine with aura evaluated higher doses not typically prescribed today, including doses exceeding 50 micrograms.^{11,12}

Additionally, this study was designed to directly assess the risk of stroke associated with use of COCs compared with alternative contraception among individuals with migraine with aura, to support clinical decision making in this population. Conversely, previous studies cited by contraception treatment guidelines have evaluated independent effects of migraine with compared with without aura on the risk of stroke, or of hormonal contraceptive use compared with nonuse among individuals with migraine broadly, without specificity to migraine with aura.^{6–8}

This study should be interpreted in the context of its limitations. Confounding by indication is

Table 3. Risk of Ischemic Stroke Over Varying Follow-Up Periods Comparing Users of Combined Oral Contraceptives With Users of Progestogen-Only Pills Among Female Individuals With Migraine With Aura in the United Kingdom

Follow-Up Time	Exposure Group	Before Weighting (Unadjusted)						After Weighting* (Adjusted)	
		No. of People	No. of Events	No. People Censored [†]	Risk/100,000 People	Crude Risk Difference/100,000 People (95% CI)	Crude RR (95% CI)	Adjusted Risk Difference/100,000 People (95% CI)	Adjusted RR (95% CI)
Primary analysis: risk of ischemic stroke, as-treated									
3 mo	COC	12,429	1	660	8.28	5.90 (−10.99 to 22.79)	3.48 (0.22–55.58)	17.62 (−21.07 to 56.30)	9.73 (0.61–155.47)
6 mo	POP	41,984	1	1,804	2.38	Ref	Ref	Ref	Ref
	COC	12,429	2	4,535	17.36	12.40 (−12.65 to 37.45)	3.50 (0.49–24.87)	22.39 (−22.23 to 67.01)	4.07 (0.50–33.35)
12 mo [‡]	POP	41,984	2	18,136	4.96	Ref	Ref	Ref	Ref
	COC	12,429	2	7,824	17.36	−0.26 (−30.85 to 30.33)	0.99 (0.17–5.68)	9.53 (−38.65 to 57.71)	1.47 (0.24–8.90)
18 mo	POP	41,984	4	28,486	17.62	Ref	Ref	Ref	Ref
	COC	12,429	2	9,285	17.36	−10.48 (−47.04 to 26.08)	0.62 (0.11–3.42)	−0.95 (−53.33 to 51.43)	0.97 (0.17–5.56)
	POP	41,984	5	33,403	27.84	Ref	Ref	Ref	Ref
Sensitivity analysis: risk of ischemic stroke—intent-to-treat									
3 mo	COC	12,429	1	199	8.11	5.73 (−10.84 to 22.30)	3.41 (0.21–54.45)	16.51 (−20.12 to 53.13)	9.00 (0.56–143.80)
6 mo	POP	41,984	1	1,353	2.38	Ref	Ref	Ref	Ref
	COC	12,429	2	447	16.42	11.54 (−12.21 to 35.29)	3.36 (0.47–23.87)	20.24 (−21.69 to 62.16)	3.76 (0.46–30.93)
12 mo [‡]	POP	41,984	2	2,834	4.89	Ref	Ref	Ref	Ref
	COC	12,429	2	970	16.42	0.79 (−25.20 to 26.77)	1.05 (0.21–5.21)	10.36 (−32.70 to 53.42)	1.60 (0.29–8.77)
18 mo	POP	41,984	6	5,785	15.64	Ref	Ref	Ref	Ref
	COC	12,429	2	1,465	16.42	−2.13 (−28.74 to 24.47)	0.89 (0.18–4.26)	7.89 (−35.45 to 51.22)	1.40 (0.26–7.42)
	POP	41,984	7	8,454	18.56	Ref	Ref	Ref	Ref

RR, risk ratio; COC, combined oral contraceptive; POP, progestogen-only pill; Ref, reference.

* All weighted analyses were weighted by the inverse probability of treatment. As-treated analyses (including the primary analysis) were additionally weighted by inverse probability of censoring weights.

[†] In as-treated analyses (including the primary analysis), individuals were censored on the occurrence of stroke, end of available data, maximum follow-up (1 year), death, discontinuation of index medication, or switching from index medication. In intent-to-treat analyses, individuals were not censored on discontinuation or switching.

[‡] The primary follow-up period under study is 12 months; results for secondary follow-up periods are also reported.

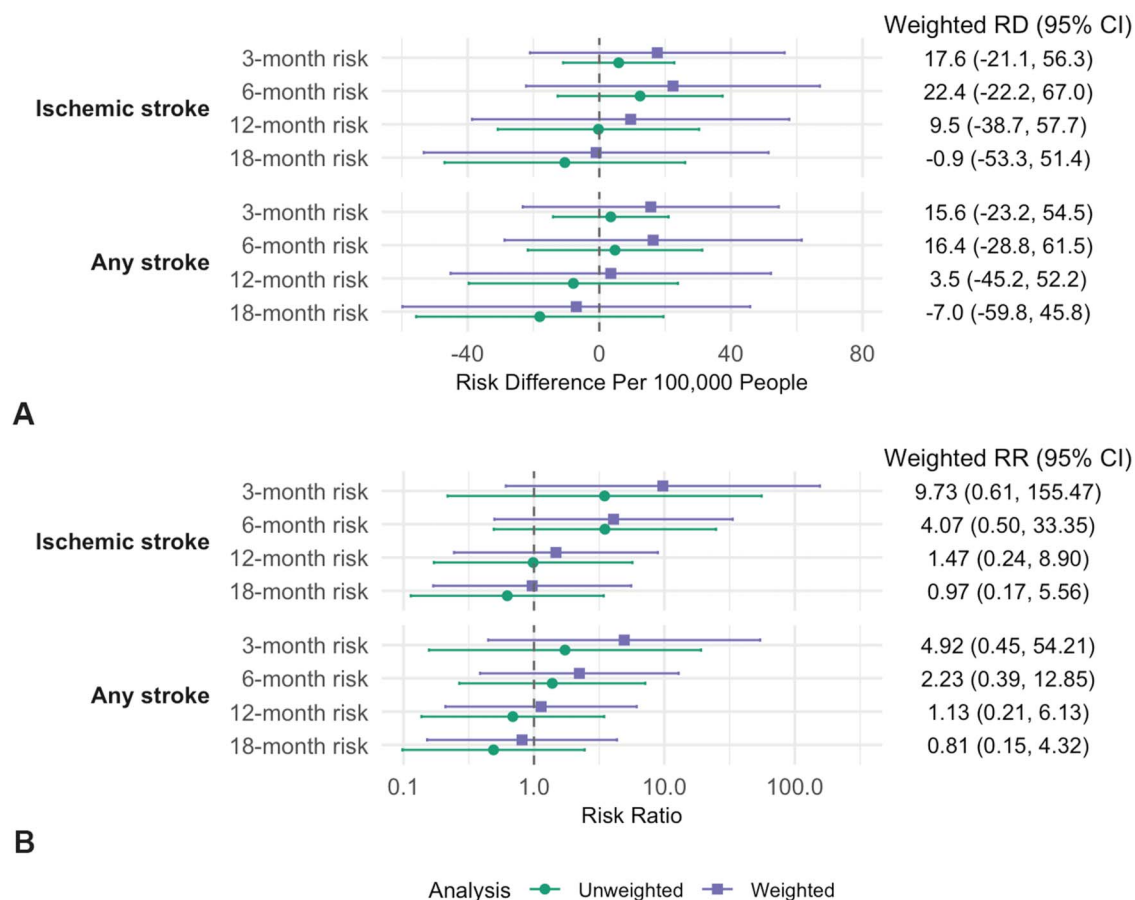


Fig. 2. Risk differences (**A**) and risk ratios (**B**) for ischemic and any stroke comparing initiators of combined oral contraceptives with initiators of progestogen-only pills among female individuals with migraine with aura in the United Kingdom. All weighted analyses were weighted by the inverse probability of treatment. As-treated analyses (including the primary analysis) were additionally weighted by inverse probability of censoring. In as-treated analyses (including the primary analysis), individuals were censored on the occurrence of stroke, end of available data, maximum follow-up, death, or discontinuation of or switching from index medication. In intent-to-treat analyses, individuals were not censored on discontinuation or switching. Any stroke (secondary outcome) includes ischemic, hemorrhagic, and unspecified stroke. RD, risk difference; RR, risk ratio.

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anticipated in these data given that some individuals at higher risk for stroke may be preferentially prescribed POPs rather than COCs. Measured confounding was addressed by exclusion of those with a history of thrombotic disease at baseline and by application of treatment weights, after which the treatment groups showed balance of baseline characteristics. Due to sparsity of outcome events, evaluation of subgroups of interest such as those based on age, dose, and migraine treatment was not feasible.

Residual confounding is possible in these data, for example, as a result of incomplete capture of some important covariates such as BMI (14% missing) and smoking status (6% missing). Missingness was addressed by using multiple imputation, which has been

shown to mitigate bias compared with complete case analysis.²⁸ Additionally, migraine frequency, which is associated with risk of stroke,^{13,29} is not well captured in these data. Thus, migraine treatment utilization was used as a proxy for migraine frequency.

Diagnoses captured in these records, including migraine with aura, were identified on the basis of diagnoses recorded by practitioners. As such, diagnoses may be imperfect or inconsistent, and some individuals with migraine with aura may not have been included in this study if not diagnosed as such by their general practitioner. Although there was insufficient information available to determine alignment of migraine diagnoses with standardized diagnostic criteria such as the International Classification of

Headache Disorders, these diagnoses were made by general practitioners and reflect everyday practice.

Although this study population is considered representative of female individuals with migraine with aura in the United Kingdom, generalizability to other populations may be limited. For example, participants were predominantly White, so potential differences in stroke risk by race or ethnicity (as observed in U.S. populations)³⁰ were not well captured by the present study data. Finally, the estimates in this study were imprecise. Specifically, CIs were wide due to the low absolute risk of stroke in this population overall.

In this cohort study of female individuals with migraine with aura without a history of thrombotic disease, those using COCs had an excess of one stroke per 10,493 people in the year after contraceptive initiation compared with those using POPs. This increase in risk is comparable with that observed among general populations that included individuals without migraine, suggesting that, relative to POPs, COCs may not confer substantially more risk for individuals with migraine with aura specifically compared with a general population. Importantly, estimates were imprecisely measured because of the low absolute risk of stroke in this population regardless of contraceptive type. Additional research into the safety of modern-day contraceptives among individuals with migraine with aura should improve the precision of these clinically relevant comparisons to better support patient and clinician decision making. However, these data provide reassurance that the risk of stroke in this patient population is not greatly increased by the use of COCs.

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